

RESEARCH ARTICLE OPEN ACCESS

Unveiling Cancer: A Data-Driven Approach for Early Identification and Prediction Using F-RUS-RF Model

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Received: 26 December 2023 | **Revised:** 23 September 2024 | **Accepted:** 31 October 2024

Funding: This work was supported by Health Department, Blekinge Tekniska Högskola.

Keywords: artificial intelligence | cancer | convolutional neural network | deep learning | medical imaging

ABSTRACT

Globally, cancer is the second-leading cause of death after cardiovascular disease. To improve survival rates, risk factors and cancer predictors must be identified early. From the literature, researchers have developed several kinds of machine learning-based diagnostic systems for early cancer prediction. This study presented a diagnostic system that can identify the risk factors linked to the onset of cancer in order to anticipate cancer early. The newly constructed diagnostic system consists of two modules: the first module relies on a statistical F-score method to rank the variables in the dataset, and the second module deploys the random forest (RF) model for classification. Using a genetic algorithm, the hyperparameters of the RF model were optimized for improved accuracy. A dataset including 10 765 samples with 74 variables per sample was gathered from the Swedish National Study on Aging and Care (SNAC). The acquired dataset has a bias issue due to the extreme imbalance between the classes. In order to address this issue and prevent bias in the newly constructed model, we balanced the classes using a random undersampling strategy. The model's components are integrated into a single unit called F-RUS-RF. With a sensitivity of 92.25% and a specificity of 85.14%, the F-RUS-RF model achieved the highest accuracy of 86.15%, utilizing only six highly ranked variables according to the statistical F-score approach. We can lower the incidence of cancer in the aging population by addressing the risk factors for cancer that the F-RUS-RF model found.

1 | Introduction

The word cancer first appeared in Greek as tumor or cancer. In medicine, the term cancer was first used in the 1600s to describe invasive or spreading cells that could damage others [1]. The process by which cancer cells extend to other organs after growing uncontrollably in one area of the body is known as metastasis [2]. It is a hereditary disease caused by genetic changes that control the functions of our cells, mainly how

they divide and spread. As tumor cells continue to increase, new changes occur. In short, cancer cells have more genetic differences compared to healthy cells, such as DNA mutations [3]. Benign cells do not spread to other parts of the body, unlike malignant cells, which do and are considered more destructive. Because of high mortality and recurrence rates, cancer treatment is time-consuming and costly. Early and accurate diagnosis is required to improve the survival rate of cancer patients.

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Cancer remains the second leading cause of cardiovascular disease worldwide [4], despite the availability of medications that could increase patients' life expectancy. Cancer affects children, adolescents, and young adults, but most people receive a cancer diagnosis in their fifth or sixth decade of life [5]. More than any other age group, the population of people over 65 is increasing worldwide [6]. The likelihood of developing cancer increases dramatically with age. Nearly 60% of cancer cases occur in people over the age of 65. In addition, about 70% of cancer-related deaths occur at age 65. Cancer is a disease that comes with age [7]. Given the increase in cancer incidence and quality of life in the aging population, an innovative approach to diagnosing, treating, and surviving older cancer patients is essential.

Early cancer detection is critical because it increases survival rates [8]. Older people diagnosed with cancer are more likely to experience morbidity and mortality from cancer surgery and to be unable to tolerate chemotherapy and radiation [9]. For this reason, caring for older people with cancer symptoms is difficult [10]. An objective strategy for detecting and treating cancer symptoms in older adults is needed. The importance of early cancer detection in older patients must be weighed against the prognosis of the disease, the prospect of successful and well-tolerated treatment, the presence of other conditions, and patient preferences. Some older people value quality over quantity [11], are less likely to have their cancer symptoms investigated, and are willing to accept a higher probability that cancer [12].

The goal of machine learning (ML), a subfield of artificial intelligence, is to find patterns in data that are helpful for predictions. Researchers developed several methods and approaches based on ML techniques for earlier prediction of various diseases such as heart failure [13] Parkinson's disease [14], hepatitis [15], dementia [16–18], cancer [19–22] using medical data. The healthcare system generates a massive amount of patient data, that is, medical history, physical examination, biochemical tests, lifestyle, and demographic data [23–25]. Using ML and deep learning (DL) methods, researchers have used these data to analyze disease predictions. ML and DL-based techniques can provide good accuracy and a non-invasive method for early cancer detection.

Based on the algorithms of ML, researchers have presented several automatic diagnostic systems for cancer detection. Park et al. designed a graph-based semisupervised learning algorithm for breast cancer prediction using a 5-fold cross-validation scheme. Their proposed ML model achieved accuracy of 71% [26]. Seng et al. used support vector machines (SVM) to detect cervical cancer. They used a hold-out validation scheme to validate the performance of their proposed model. The dataset they used to test the effectiveness of their proposed ML model consisted of 168 samples. SVM achieved the reported accuracy of 68% [27]. Sudharani et al. demonstrated the automatic segmentation method that separates non-enhancing brain tumors from healthy tissue by locating the tumor position in the brain and providing a complete statistical analysis of the tumor using MR images [28]. Suberi et al. [29] developed an image-based, computer-aided approach for cancer immunotherapy. The suggested system's

overall precision, recall, and accuracy are 64.9%, 71.7%, and 87.8%, respectively. Rahman et al. [30] tested several ML methods, such as logistic regression (LR), linear discriminant analysis (LDA), and SVM for oral squamous cell prediction. However, the dataset used for the experiments consisted of multiple samples, for example, only 42 oral squamous cells. Breast cancer risk variables were identified using the XGBOOST (XGB) classifier based on interrelated genetic and sociodemographic characteristics by Behravan et al. [31]. Using a cross-validation technique ($k=10$ and $k=5$), their ML-based model achieved 78% accuracy. Alsmariy et al. [32] employed a voting procedure that includes three classifiers: decision tree (DT), LR, and random forest (RF). These classifiers were used to build the classification model. The University of California, Irvine (UCI) cervical cancer risk factor dataset was acquired to test the proposed classifier. The data samples in the collected dataset were highly imbalanced. To overcome the problem of the imbalanced dataset, the synthetic minority oversampling technique (SMOTE) with the principal component analysis (PCA) approach was employed to reduce the dimensions of data that do not affect ML model accuracy. A stratified 10-fold cross-validation approach was then applied to avoid the problem of overfitting. Using routine clinical and laboratory data, the method ML was developed by Gould et al. to detect lung cancer in its early stages by Gould et al. [33]. Their proposed method achieved the highest area under the curve (AUC) of 0.86%. Liu et al. [34] developed an integrated ML method to generate an immune-related lncRNA consensus signature (IRLS). By integrating 10 ML algorithms, a consensus IRLS with 85% accuracy was developed. Most recently, Ali et al. employed an ML-based statistical analysis for identifying the pre-stage of cervical cancer [35]. Chae et al. analyzed a longitudinal study by employing hospital data on cancer patients ($n=208$) between 2008 and 2014. Recurrent neural networks (RNN) are constructed using long short-term memory (LSTM), trained on past symptom experiences, and utilized to forecast symptom trajectories in the future [36]. Giannakeas et al. used ML and electronic medical information to detect ovarian cancer early.

According to the literature review, researchers have worked extensively with ML techniques for early cancer diagnosis. They used different data modalities, such as images and patient EHRs. However, the ML models proposed for cancer detection have used small datasets, which means they suffer from biases in ML models and overfitting or lack of generalization to new data problems. In addition, this study mainly focuses on older adults' data for cancer prediction using ML. The main contributions of this study are given as follows:

- This study discusses the problem of improper validation techniques used in earlier research and promotes the use of substitute validation strategies for early prediction of cancer.
- We collect electronic health record (EHR) data from two SNAC facilities (Blekinge and Kungsholmen) for this study. The SNAC's standardized guidelines are used to harmonize the collected data.
- To resolve the problem of bias due to the imbalanced classes in the collected dataset, we employed the random under-sampling (RUS) technique for accurate results.

- We improve the set of significant variables by applying a statistical F-score technique. By efficiently removing unnecessary and redundant variables, this method produces a feature set of greater quality that may be used for classification.
- As far as we are aware, the F-RUS-RF hybrid diagnostic system is a novel approach for early-onset cancer using electronic health records.

2 | Materials and Methods

2.1 | Dataset Description

The Swedish National Study on Aging and Care (SNAC) served as the data source for this study. SNAC is a long-term consortium that collects multimodal data from Sweden's aging population to create reliable, comparable, and durable datasets for aging research [37]. SNAC was created as a multipurpose program to study the standard of health care for the aging population. SNAC includes several databases with data on various topics, including medical records, social variables, lifestyle factors, metacognitive information, and physical examinations. SNAC collected data on Swedish seniors from Blekinge, Skåne, Nordanstig, and Kungsholmen, representing urban and rural areas. This study incorporates data from metropolitan areas and is based on established standards (Blekinge and Kungsholmen). The following is the exclusion process that was applied to remove some subjects from this investigation:

- Subjects for whom the outcome field had missing data.
- Subjects whose incomplete data percentage exceeded 30%.
- Subjects who at baseline diagnosed with cancer.

After applying the exclusion criteria, we collected data from 10765 observations: 3461 from SNAC-Kungsholmen and 7304 from SNAC-Blekinge. The main goal of data collection was to integrate and harmonize SNAC data from the two sites. The dataset was standardized by following the harmonization rule for the variables. The dataset includes 3949 male observations and 6816 female observations. Only 1477 SNAC observations had cancer; all others were classified as healthy. The distribution of observations within the given dataset SNAC is shown in Figure 1, with the quantities of male and female observations,

healthy individuals, and cancer patients distinguished by several color bars. Motivated by the published research on cancer from the Swedish Cancer Register [38–40], we selected variables from SNAC databases (Blekinge and Kungsholmen). Seventy-four variables and one outcome variable (cancer) were identified from the eight categories of demographics, social, lifestyle, medical history, physical examination, biochemical tests, psychology, and evaluation of various health instruments as given in Table 1.

2.2 | Proposed Method

In this study, we used EHR to develop an automated diagnostic system to predict and identify risk factors for cancer in older adults. The newly proposed automated diagnostic system operates as a black box consisting of two main components: The first component is used to rank the variables from the entire dataset that can help predict cancer. In contrast, the second component is used to classify healthy individuals and patients by using selected variables from the first component. For ranking the significant variables from the whole dataset, we used the f1 score, while for classification, the RF was used. An overview of the operation of the proposed automatic diagnostic system for predicting cancer in older adults is given in Figure 2.

In the first phase, a statistical model is used to calculate the F-score for each selected variable according to the following formula:

The distinction between two sets of real numbers is measured by the F-score for the variable ranking model [41]. Suppose the number of instances associated with healthy observations is v_+ , and the number of instances associated with the patient group is v_- for a given dataset of T_p , $p = 1, 2, 3, \dots, n$ occurrences. In that case, the F-score of the k th variables is defined as:

$$F(k) = \frac{\left(\bar{T}_k^{(+)} - \bar{T}_k\right)^2 + \left(\bar{T}_k^{(-)} - \bar{T}_k\right)^2}{\frac{1}{(v_+ - 1)} \sum_{p=1}^{v_+} \left(\bar{T}_{p,k}^{(+)} - \bar{T}_k^{(+)}\right)^2 + \frac{1}{(v_- - 1)} \sum_{p=1}^{v_-} \left(\bar{T}_{p,k}^{(-)} - \bar{T}_k^{(-)}\right)^2} \quad (1)$$

The means of k th variables from complete datasets (positive, negative) classes are given as $\bar{T}_k$, $\bar{T}_k^+$, $\bar{T}_k^-$ respectively. Moreover, $\bar{T}_{p,k}^{(-)}$ is the k th variable of the p th negative sample. $\bar{T}_{p,k}^{(+)}$ is the k th

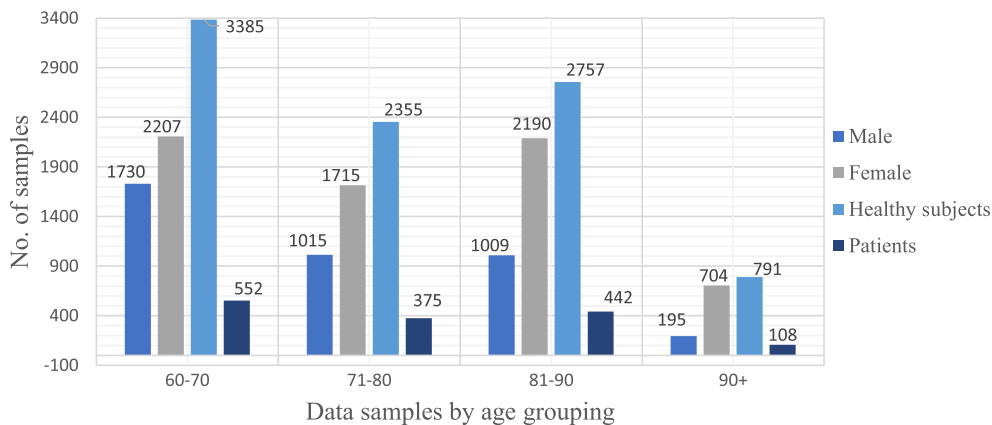
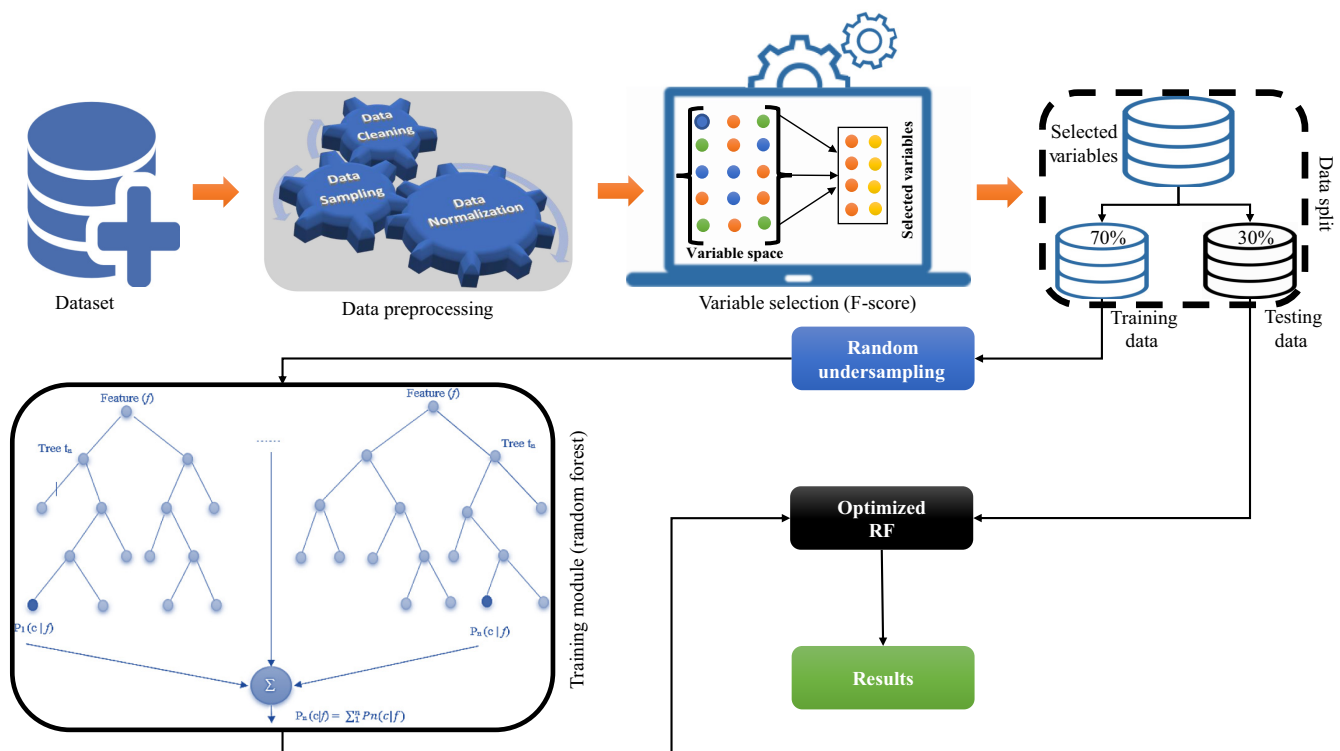


FIGURE 1 | Distribution of observations in the SNAC (Blekinge and Kungsholmen) dataset.

TABLE 1 | Description of selected variables from SNAC dataset.

Variable_category	Variables
Demographic	Gender, age
Social	Support network, education, loneliness, religious belief, social network, voluntary association, and religious activities
Lifestyle	Physical-workload, past smoker, physically demanding activities, present smoker, alcohol quantity, leisure activities, number of cigarettes a day, social activities, work status, alcohol consumption, light exercise
Medical history	High blood pressure, myocardial infarction, head trauma, arrhythmia, sleep apnea, tia/rind, diabetes type 1, stroke, thyroid disease, family history of importance, cancer, epilepsy, number of medications, diabetes type 2, parkinson's disease, other psychiatric diseases, snoring, atrial fibrillation, hip fracture, cardiovascular ischemia, heart failure, developmental disabilities, and depression
Physical examination	Blood pressure on the right arm, number of teeth, body mass index (BMI), assessment of rising from a chair, heart rate sitting, heart rate lying, hand strength in right arm in 10s interval, hand strength in left arm in a 10s interval, single-leg standing with right leg, single leg standing with left leg, pain in the last 4 weeks, dental prosthesis, feeling of safety from rising from a chair
Biochemical test	C-reactive protein analysis, hemoglobin analysis
Psychological	Sense of identity, memory loss, personality change, memory decline, abstract thinking, and memory decline 2
Health instruments	Backwards digit span test, digit span test, livingston index, sense of coherence eq5d test, instrumental activities of daily living, mini-mental state examination, activities of daily living, comprehensive psychopathological rating scale, physical composite score of the sf-12 health survey, clock drawing test, mental composite score of the sf-12 health survey

Note: TIA/RIND, transient ischemic attack/reversible ischemic neurological deficits.

**FIGURE 2** | Working of proposed diagnostic (F-RUS-RF) model.

variable of the p th positive sample. In Equation (1), the denominator denotes a variable within each of the two groups, while the numerator formulates the distinction between the positive

and negative classes. The discriminative ability of a variable is inversely correlated with its F-score. After ranking the variables using an F-score-based statistical model, an F-score

threshold must be set, that is, only variables with F-scores above the threshold are considered.

After selecting the significant variables, the data are divided into training and testing datasets for the proposed approach using holdout validation. The collected dataset consists of 10765 samples, of which 86% (9258) were classified as healthy and 14% (1507) were patients. The classes in the dataset were found to be highly imbalanced. Therefore, we applied the RUS method to solve the ML model's bias problem to balance the classes in the training data [42]. After RUS, we got an even number of samples (1200) for each health and patient class. The model was trained on a balanced dataset through RUS and tested on the original validation set so that it should not be overfitted as shown in Figure 2. Figure 3 presents the original and balanced number of samples after applying the RUS method.

The formulation of the RF model is as follows:

Ensemble classifier $\omega(\partial, \mathfrak{V})$, where $\mathfrak{V}$ are similarly distributed random vectors from RF. The most evident class from each participating tree is selected as input α . For the input sample, ω similar to the samples of the training dataset μ and σ were extracted from the F-feature space. $\mathfrak{R}$ training datasets are obtained, such as $\eta_1, \eta_2, \dots, \eta_k$ by n times random selection. Each tree in the forest is fully grown except for pruning, and the DT $\wp_1, \wp_2, \dots, \wp_k$ is generated from the corresponding training sets.

Numerous decision trees are used for categorization in the development of random forests. Two hyperparameters, such as the number of trees in the forest (E) and the (D) maximum depth of each tree, are crucial for classification in the DT [43]. In this work, we used a genetic algorithm to determine the best values for E and D , which ultimately improved the functionality of the RF model. As a result, a RF model is built, new sample data is added, and decision trees are used to determine the nature of the new sample. Ultimately, each tree in the RF casts

a vote to determine how to classify the delivered data sample. The trees in a RF are created by bootstrapping resampling onto the training data. Bootstrap is used for model integration with a replacement approach since it is an easy and dependable way [44]. Using bootstrap random sampling, a defined number of samples are taken from the training set. Following sampling, the number of samples is returned to the training set. The extracted data are used to generate a new batch of bootstrap samples, and the collected samples may be sampled again after they have been returned to the training set. Before returning to the training set, acquired samples would likely be sampled again.

For example, a φ sample is drawn at random. The probability of collecting each time is expressed as $1/\varphi$ and $(1-1/\varphi)$. For a T times sample, the probability of the sample is given as $(1-1/\varphi)^T$, where T converges to ∞ and $(1-1/\varphi)^T$ converges to $1/e = 0.368$. A mirror sample is already available, and the new samples are used in 1 out of 3 cases. The data that is missed during the data extraction process is called out-of-bag (OOB) error and is represented as follows:

$$\text{OBE} = \frac{\mathbb{T}}{\text{OBB}} \quad (2)$$

In Equation (2), $\mathbb{T}$ denotes the error of testing OBB data, and OBB represents OOB.

The CART algorithm uses the Gini index to represent the degree of impurity of the model and to help build a decision tree. A model has low impurity if the Gini index value is low. For a classification problem where the probability of the n th category is ξ for N classes, the Gini index formulation is as follows:

$$\text{Gini}(H) = \sum_{n=1}^n \xi_n (1 - \xi_n) = 1 - \sum_{n=1}^n \xi_n^2 \quad (3)$$

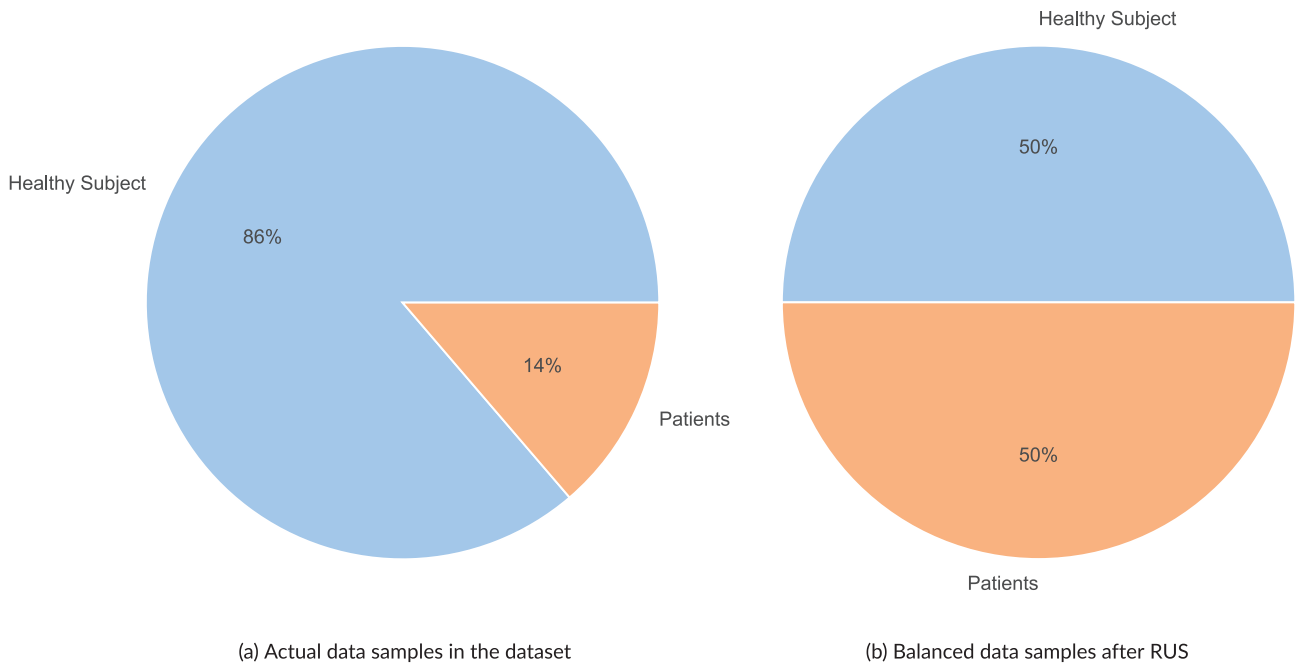


FIGURE 3 | Data samples before and after RUS.

Decision trees use the Gini Index to select variables, and their mathematical computation is as follows:

$$\Delta\text{Gini}(G) = \text{Gini}(H) - \text{Gini}_q(H') \quad (4)$$

Decision trees can be used to compare the overfitting of models. Therefore, pre- and post-pruning techniques are used to reduce the rate of overfitting. In addition, pre-pruning causes the DT to evolve quickly, while post-pruning can lead to better results. A DT was created without pruning to obtain better results.

Significant variables from the dataset, identified using the F score, were entered into the RF model for cancer classification after the RF was designed and optimized. The proposed model was trained using 70% of the dataset based on a holdout validation approach. While 30% of the data are used to evaluate the model's predictions, by eliminating the problems of overfitting the model and unbalanced classes in the dataset, the newly proposed model has shown improved performance and achieved the highest accuracy for unbiased cancer prediction.

2.3 | Validation and Evaluation

The holdout validation method is often used in the literature as a standard to evaluate the performance of ML-based diagnostic systems [45–47]. A holdout validation technique splits a dataset into two; one segment is used for training, while the other is used to test the proposed ML model. The dataset is split using 30% for testing and 70% for training the ML model. Therefore, our experiments used the above data-splitting criteria to train and evaluate the proposed approach.

After splitting the dataset, we use evaluation metrics to assess how well the proposed model compares to other state-of-the-art ML models for cancer prediction. Accuracy, specificity, sensitivity, f1score, Matthew's correlation coefficient (MCC), and AUC using a receiver operator characteristic curve (ROC) and 95% confidence interval are the evaluation metrics used to assess the proposed model. The mathematically provided evaluation metrics are given as follows:

$$\text{Accuracy} = \frac{\tau p + \tau n}{\tau p + \tau n + fp + fn} \quad (5)$$

where τp stand for the number of true positive, fp represents number of false positive, τn denotes the number of true negative, and fn represents the number of false negative.

$$\text{Sensitivity} = \frac{\tau p}{\tau p + fn} \quad (6)$$

$$\text{Specificity} = \frac{\tau n}{\tau n + fp} \quad (7)$$

$$\text{F1_score} = \frac{2\tau p}{2\tau p + fp + fn} \quad (8)$$

$$\text{MCC} = \frac{\tau p \times \tau n - fp \times fn}{\sqrt{(\tau p + fp)(\tau p + fn)(\tau n + fp)(\tau n + fn)}} \quad (9)$$

A statistical analysis of a binary classification problem is performed. The F-measure is also referred to as the *F1-score*. The F-score indicates a value between 0 and 1, where 1 shows good predictions, and 0 indicates poor predictions. MCC is used to determine the accuracy of a test. MCC has a value range between 1 and -1 , where 1 stands for the best and -1 for the worst prediction. Statisticians and data analysts utilize confidence intervals to determine the statistical validity of their estimates, findings, or predictions. It is impossible to conclude that a result through testing or experimentation is a consequence of a particular factor instead of coincidence if a confidence interval contains the number zero (or some other null hypothesis). The mathematical formulation for CI is based on sample mean ($\bar{X}$), Z-score (Z) and standard deviation(σ) are given as follows:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)}{N}} \quad (10)$$

$$Z = \frac{x - \mu}{\sigma} \quad (11)$$

From Equation (10) and Equation (11), we can calculate the CI as follows:

$$\text{CI} = \bar{X} \pm Z \left(\frac{\sigma}{\sqrt{n}} \right) \quad (12)$$

3 | Results

Three different types of experiments were conducted to thoroughly evaluate the effectiveness and performance of models from ML and a recently proposed model (F-RUS-RF) for predicting cancer and identifying causes. In the first experiment, the ML models' effectiveness was tested on balanced data using all variables in the dataset. In the second experiment, we used an F-score based variable selection module to select the most important variables and an RF model to perform the classification task. The hyperparameters of RF were optimized using a genetic algorithm. In the third experiment, we used the same variable selection module based on F-scores and other ML models for the classification task. All computations were performed on a 2.40GHz Intel (R) Core (TM) i7-1185G7 CPU running Windows 10 64-bit. Python software was used to perform each experiment.

3.1 | Experiment 01: Performance of ML Models on Balanced Data

We used Python to build several ML models, including Naive Bayes (NB), XGB, LR, DT, RF, K-nearest neighbors (KNN), and support vector machine (SVM) with an RBF kernel. The efficacy of the constructed ML models was tested using all variables in the dataset (74). Significantly, the RUS approach was used to balance the classes in the dataset. In this experiment, a holdout validation scheme was used where 70% of the data was used to train the ML models, and 30% was used for testing. As shown in Table 2, the effectiveness of the ML models was evaluated using several evaluation metrics, including accuracy, sensitivity, specificity, F1 score,

TABLE 2 | The performance of ML models on balanced dataset.

Model	Train _{Acc}	Test _{Acc}	Sens.	Spec.	F1_score	MCC	95%CI
NB	62.70	50.74	72.64	25.18	42.00	0.2067	0.48–0.52
LR	69.75	54.20	88.16	20.45	54.00	0.3345	0.52–0.56
KNN	59.58	53.41	87.92	13.77	53.00	0.2526	0.51–0.55
XGB	95.80	62.03	90.18	38.19	62.00	0.4061	0.59–0.64
Adaboost	83.15	62.87	85.00	56.22	65.00	0.4284	0.59–0.64
DT	99.00	55.82	79.10	50.32	56.00	0.3871	0.52–0.56
RF	95.86	65.08	91.22	66.14	65.00	0.4412	0.62–0.66
SVM _{rbf}	76.50	54.24	88.57	49.58	54.00	0.3286	0.52–0.56

Abbreviations: CI, confidence interval; MCC, Matthews correlation coefficient.

MCC, 95% confidence intervals, and ROCs. The performance of the ML models based on the ROC curve can be seen in Figure 4.

3.2 | Experiment 02: Performance of the Proposed F-RUS-RF Model

In this experiment, we evaluated the performance of the newly proposed F-RUS-RF model for cancer prediction in older adults. The collected dataset from SNAC contains a total of 74 variables. To select the most significant variables from the dataset, we applied an F-score statistical model to rank the variables. It is worth noting that the classes in the collected dataset needed to be more balanced. To avoid the problem of bias in the ML model, we used the technique RUS. After balancing the classes in the dataset, we used RF for the classification task. The hyperparameters of the RF model were optimized using genetic algorithms that helped us determine the optimal depth of the tree (D) and its edges (E). Different sets of top selected variables from the F-score statistical model were fed into an optimized RF model for classification. The performance of the newly created F-RUS-RF model was evaluated using various evaluation metrics such as accuracy, sensitivity, specificity, MCC, and 95% CI using a holdout validation scheme in which 70% of the data were used to train the ML models, and 30% were used to test the proposed F-RUS-RF model. Table 3 provides an overview of the performance of the proposed F-RUS-RF model along with the hyperparameters from RF, where the proposed F-RUS-RF model achieved the highest accuracy of 86.15% by using only the top 6 ranked variables from the dataset. The best-selected variables from the F-score test were age, personal care, arrhythmia, head trauma, mental health issue, and dementia.

Furthermore, the performance of the proposed F-RUS-RF model is also compared with the conventional RF model. The conventional RF model used the top six variables which are ranked by F-score module as in the proposed model. To evaluate the performance of the traditional RF model and the newly proposed F-RUS-RF model, we employed the receiver operator curve (ROC). ROC is a visual graph that shows when the discrimination threshold of a binary classification system is adjusted and how well it can diagnose problems. The true positive rate (TPR) and false positive rate (FPR) are plotted against each other at different thresholds to create the ROC curve. In

addition to sensitivity, recall, and probability of detection, the true-positive rate has several names. The FPR can be calculated as (1 specificity), another name for the probability of a false alarm. The AUC is used to measure the performance of the ML model with ROC. A graph with a larger AUC is considered more accurate. In addition, we used a cross-validation scheme to test the accuracy of the newly proposed F-RUS-RF model [48]. We used the k-fold cross-validation technique for the ROC curve, choosing the value $k = 5$. Cross-validation uses multiple subsets of data to evaluate and train a model over numerous rounds. It is typically applied when the goal is prediction, and one wants to estimate how well a predictive model will perform in real-world situations. Figure 5 shows the performance comparison between the conventional RF model and the newly developed F-RUS-RF model. Also, Figure 5b shows that the recently proposed model has an AUC of 86% compared to the conventional RF model, which has an AUC of 84% as shown in Figure 5a. Hence, the proposed model (F-RUS-RF) improved the performance of the conventional RF model by 2.38% because the hyperparameters of RF were fine-tuned using a genetic algorithm in the proposed model.

3.3 | Experiment 03: Performance of ML Models Based on F-Score Variable Selection

We tested the performance of other modern ML models based on variable selection methods with f-scoring, such as NB, LR, XGB, decision trees (DT), Adaboost, and SVM with different kernel functions (Linear, BRF, Poly, Sigmoid). The dataset contains 74 variables, and we used the F-score approach to rank the variables so that high-scoring variables can be identified to predict cancer in older adults. All variables are ranked according to their relevance as given in Figure 6a. In contrast, the top 11 selected variables selected by the proposed method (F-score) can be shown in Figure 6b. The number of variables (features) is shown on the x-axis, and the ranking of the variables(features) is shown on the y-axis in Figure 6. Each variable was ranked, and the top 11 variables with values greater than five were selected from the 74 variables. After choosing the top-ranked variables, we decided on the most advanced ML models to evaluate their performance on the best features. The hyperparameters of the models from ML were optimized using an exhaustive grid search algorithm. We applied

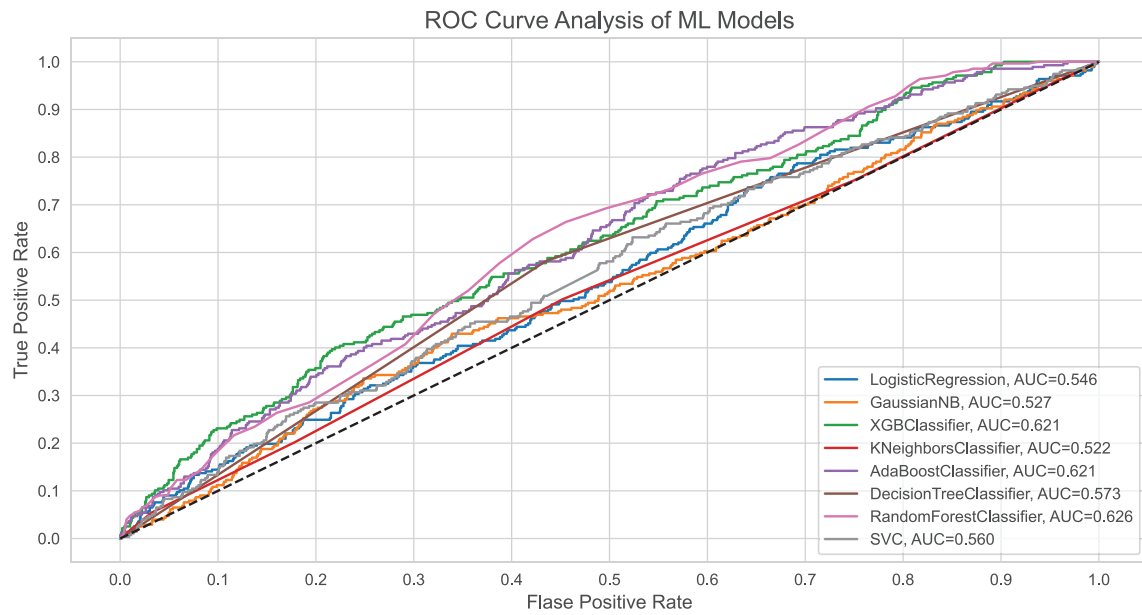


FIGURE 4 | Performance of ML models based on the ROC curve.

TABLE 3 | The performance of the proposed F-RUS-RF model.

S_Var.	Hyp.	Train _{Acc}	Test _{Acc}	Sens.	Spec.	MCC	95%CI
11	$D=3, E=1$	65.21	59.69	75.54	44.23	0.3012	0.58–0.62
10	$D=1, E=4$	62.35	58.00	69.13	42.75	0.2887	0.56–0.60
12	$D=6, E=1$	71.55	66.15	70.00	55.21	0.3587	0.64–0.68
08	$D=4, E=2$	79.00	73.44	78.55	63.34	0.3799	0.72–0.76
02	$D=2, E=6$	75.45	70.12	68.91	59.00	0.3115	0.68–0.72
06	$D=8, E=2$	85.23	80.00	90.00	75.41	0.4087	0.78–0.82
05	$D=15, E=10$	90.78	85.77	89.66	70.98	0.4233	0.82–0.86
09	$D=12, E=3$	88.53	86.21	90.36	80.00	0.4452	0.84–0.88
06	$D=15, E=10$	91.00	86.15	92.25	85.14	0.4511	0.85–0.89
12	$D=15, E=8$	90.55	85.00	95.00	83.22	0.4464	0.84–0.88
04	$D=05, E=2$	84.00	82.75	84.00	55.81	0.4178	0.80–0.84

Abbreviations: Hyp, hyperparameters; S_Var, selected variables.

a holdout validation scheme using 70% of the data to train the ML models and 30% to test the ML models. The performance of the ML models can be seen in Table 4 using various evaluation metrics such as accuracy, sensitivity, specificity, MCC, and 95% CI.

4 | Discussion

In this work, we presented the F-RUS-RF model for cancer prediction and identified the risk factors for cancer development in older adults. We used 74 variables from the SNAC dataset related to age-related demographic characteristics, social factors, lifestyle, medical history, biochemical tests, physical examinations, psychological assessments, and various health assessments. Three types of experiments were conducted. In this work, we

presented the F-RUS-RF model for cancer prediction and identified the risk factors for cancer development in older adults. We used 74 variables from the SNAC dataset related to age-related demographic characteristics, social factors, lifestyle, medical history, biochemical tests, physical examinations, psychological assessments, and various health assessments. Three types of experiments were conducted. In our second experiment, we proposed F-RUS-RF, a hybrid ML model that works as a black box. The proposed F-RUS-RF model consists of two modules. The first module of the model works as a variable selection module in which a statistical F-score model was developed to rank the variables based on their significance for cancer detection. We used the RF model for this task rather than the second model, a classifier. The hyperparameters of the RF model were fine-tuned using a genetic algorithm. Before classification, it

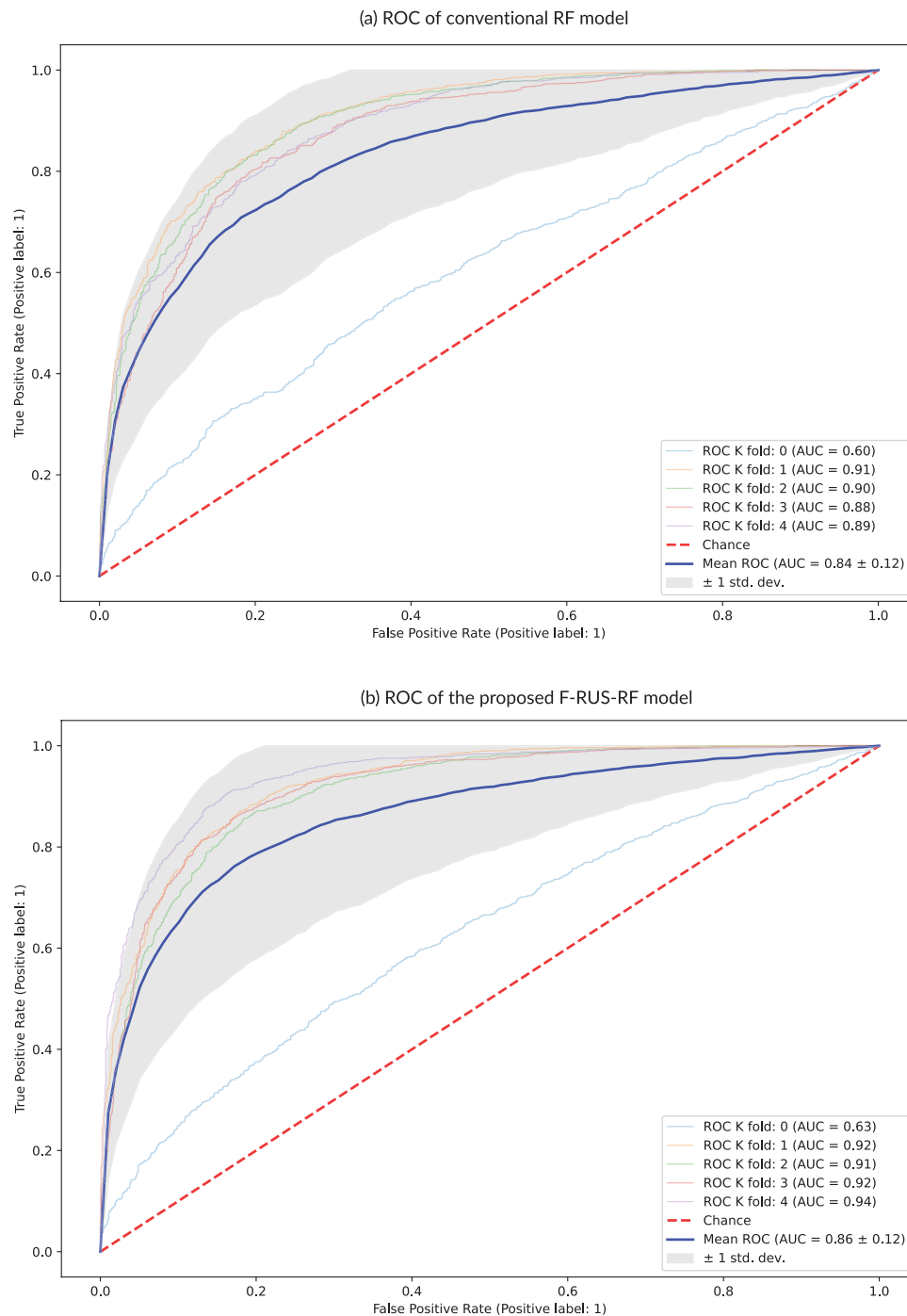


FIGURE 5 | ROC comparison of conventional RF with proposed F-RUS-RF model.

was found that the presence of classes in the dataset needed to be more balanced; to overcome this problem, we used the RUS technique.

Undersampling methods attempt to balance the data by identifying representative examples from the majority classes to minimize the data while maintaining the quality of the original data. In contrast, oversampling methods support a balanced dataset by creating new synthetic instances from the minority class. It could be questionable and unethical to create artificial, unrealistic examples of medical data and other data related to specific individuals. Most oversampling techniques also employ

mathematical techniques to generate new data, including means, standard deviations, and other metrics. Such methods are not relevant for mixed and partial data. Since oversampling methods have more instances in the dataset, the execution time of the decision algorithms [49] increases. In addition, the main drawback of oversampling is that it increases the likelihood of overfitting by using identical replicas of previous instances. It is incredibly typical for a learner to create a classification rule for a single duplicated instance when oversampling [50]. Therefore, we used the RUS method in our proposed model to avoid the problem of overfitting the model. In this study, we use the Python software package and the Imbalance learning package to develop

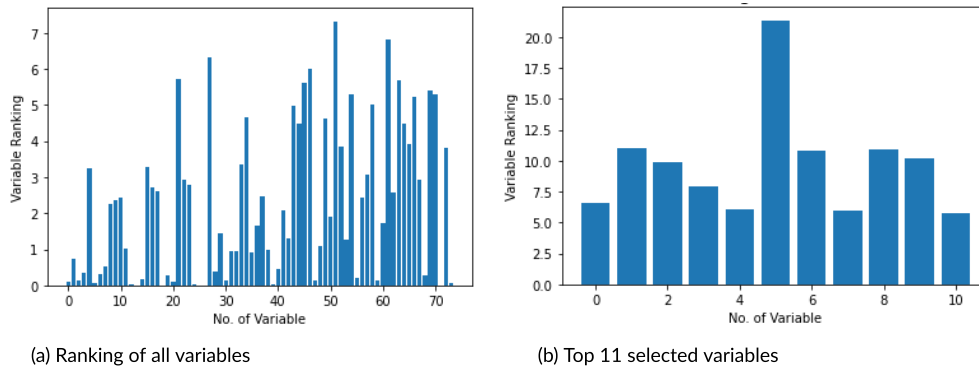


FIGURE 6 | Ranking of all variables along with top 11 selected variables by F1-score.

TABLE 4 | The performance of ML models using F1-score variable selection method.

Model	Hyp.	Train _{Acc}	Test _{Acc}	Sens	Spec	MCC	95%CI
NB	Var = 1.0	60.10	56.68	88.85	15.59	0.2815	0.55–0.58
LR	C:0.01, $P=12$	54.49	53.19	89.01	15.42	0.2584	0.52–0.55
KNN	$K=2$	86.55	70.65	86.90	13.08	0.3054	0.69–0.72
XGB	$lr=0.01$, $G=0.5$	80.54	76.53	87.40	15.89	0.3578	0.75–0.77
DT	$D=03$, Leaf = 1	89.00	82.95	65.48	17.70	0.3284	0.80–0.84
Adaboost	$lr=0.97$, $E=20$	75.00	66.52	72.61	43.53	0.2874	0.64–0.68
SVM_rbf	$C=100$, $G=1$	81.53	72.46	86.99	13.28	0.2941	0.61–0.64
SVM_lin.	$C=0.1$, $G=1$	79.50	73.00	89.00	79.01	0.3023	0.71–0.74
SVM_sig.	$C=0.1$, $G=0.01$	72.14	60.58	85.30	69.24	0.2861	0.58–0.62
SVM_poly.	$C=10$, $G=0.01$	65.46	50.98	70.51	48.97	0.2488	0.48–0.52

Abbreviation: Hyp, hyperparameters.

TABLE 5 | Top ranked selected variables by proposed model (F-RUS-RF).

S_Var. code	Description
Age	Age
E134A	Do you feel safe (rising from chair)
E66	Can you handle your personal care (independence)
K45	Arrhythmia
K50	Head trauma with loss of consciousness/syncope
K90	Mental health issues
K91	Dementia
L94A	Blood pressure systolic
DG239	DIGIT SPAN: forward longest span
DG240	DIGIT SPAN: backward longest span
O199	CPRS: inability to feel

Abbreviations: CPRS, Conners' Parent Rating Scale-Revised; S_Var, Selected Variables.

the approach RUS [49, 51, 52]. The second phase of the process involves the development of the RF model using the balanced training data obtained through the RUS technique.

The performance of the newly proposed F-RUS-RF model can be seen in Table 3. The proposed F-RUS-RF model achieved the best accuracy of 86.15% using only six variables (age, personal care, arrhythmia, head trauma, mental health issue, and dementia) from the dataset. Age is the most critical factor contributing to cancer development in older adults. Furthermore, we evaluated the performance of the conventional RF model on the balanced dataset using the most significant variables from the F-score statistical model. We employed the ROC curve to compare the performance evaluation of the newly developed F-RUS-RF model with the conventional RF model. From Figure 5, it can be seen that the newly constructed F-RUS-RF model outperforms the traditional RF model. In the third experiment, we used the F-score statistical model to rank all variables in the dataset, as shown in Figure 6a. After ranking all variables, the top 11 ranked variables were selected using the F-score statistical model as shown in Figure 6b. After selecting the highest-scoring variables, several state-of-the-art ML models were applied to classify cancer in older adults. According to Table 4, the DT had obtained the highest accuracy of 82%. The best-selected

TABLE 6 | Comparison of the proposed (F-RUS-RF) model with previously proposed ML models.

Study	Years	Model	Validation scheme	Accuracy (%)
A. N. Richter & T.M. Khoshgoftaar [53]	2018	RF	Cross-validation	80.70
A. A. Ballin et al. [54]	2019	DNN	Cross-validation	78.00
Z. Zeng et al. [55]	2019	SVM	Cross-validation	78.00
Z. Zeng et al. [55]	2019	CNN	Cross-validation	79.00
L. Appelbaum et al. [56]	2020	NN	Cross-validation	84.00
O. Bardhi & B. G. Zapiain [57]	2021	KNN	Cross-validation	83.00
I. Kaur et al. [58]	2021	Ensemble	Cross-validation	76.40
Y. Meng et al. [59]	2021	DNN	Cross-validation	76.00
M. Botlagunta et al. [60]	2023	DT	Cross-validation	83.00
C. Jeng-Wen et al. [61]	2024	LGBM	Holdout	83.00
Current study	2024	F-RUS-RF	Cross-validation	86.15

Abbreviation: LGBM, Light Gradient Boosting Machine.

variables by the F-score statistical model can be depicted from Table 5. In addition, the performance of the newly proposed F-RUS-RF model has been compared with 10 ML models that the researcher previously disclosed. The selected ML models used electronic health records to predict cancer. Table 6 shows an overview of the proposed model F-RUS-RF's performance with the 10 ML models.

Although the constructed F-RUS-RF model exhibits encouraging accuracy rates, it is uncertain if it can be applied to larger populations or various datasets. When the model is used with datasets from different locations or demographics, its performance may differ. The study ranks variables and determines the most influential ones using a statistical F-score method; nonetheless, the variables considered may not adequately reflect the complexity of cancer risk factors. The analysis may not have incorporated all pertinent features, which could have an impact on the model's ability to predict outcomes.

Future research should confirm the F-RUS-RF model's performance on independent datasets from a variety of demographics and healthcare settings in order to evaluate the model's ability to generalize. This would shed light on its adaptability and efficacy in various situations. Undertaking longitudinal research may provide a more profound comprehension of the time correlations between identified risk variables and the initial onset of cancer. This would make it possible to develop predictive models that take into consideration how risk factors change dynamically over time, improving early detection and intervention techniques. The predictive capacity of cancer diagnostic systems may be improved by combining multi-omics data, such as proteomics, metabolomics, and genomes. Combining several data modalities could reveal new biomarkers and processes linked to the emergence of cancer, producing predictive models that are more precise and thorough. Enhancing the interpretability of ML models such as the F-RUS-RF is essential to their application in clinical settings. In order to make model predictions easier to integrate into clinical decision-making processes, future research could

concentrate on creating methods for clearly and concisely explaining them.

5 | Conclusion

In this study, we proposed a hybrid system (F-RUS-F) that used EHR data to identify risk factors for cancer in order to predict cancer. The SNAC provided the dataset, which consists of 74 variables from eight categories: medical history, physical examination, biochemical testing, psychological data, lifestyle, medical demographics, and health instruments. The first component of the proposed system ranked the most significant variables from the dataset that help in predicting cancer, whereas the second component used selected variables for detecting cancer in older adults. Several metrics were used to assess the proposed system, including MCC, 95% CI, accuracy, sensitivity, specificity, and AUC, by utilizing the ROC curve. Based on the top six identified variables (age, personal care, arrhythmia, head trauma, mental health condition, and dementia), the F-RUS-RF model's highest accuracy for cancer prediction was 86.15%. Our research will ultimately lead to better health care and a higher standard of living by improving the detection and prevention of cancer in older adults. For future work, our focus will be on designing and developing robust methods based on DL methods while using a multimodal dataset. In our next work, we will use multimodality datasets to design and develop robust methods for cancer prediction based on DL techniques.

Author Contributions

Conceptualization: Ashir Javeed and Peter Anderberg. Data curation: Johan Sanmartin Berglund. Formal analysis: Ashir Javeed and Muhammad Asim Saleem. Writing – original draft: Ashir Javeed, Peter Anderberg and Ahmad Nauman Ghazi. Methodology: Ashir Javeed. Writing – review and editing: Ahmad Nauman Ghazi, Ashir Javeed and Muhammad Asim Saleem. All authors have read and agreed to the published version of the manuscript.

Acknowledgements

This research is being carried out with the assistance of the Swedish National Study on Aging and Care. SNAC contributed the electronic health record (EHR) data for this study. This study is being conducted at the Blekinge Institute of Technology's Health Department.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data can be accessed on demand from the SNAC data center.

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